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MILITARY SPECIFICATION

CARTRIDGE, 106MM, HEAT, M344A1, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading assembling and packing for one type of anti-tank cartridge designated as Cartridge, 106-MM, HEAT, M344A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-R-11470 — Radiographic Inspection, Qualification of Equipment, Operators and Procedures.

MIL-R-11471 — Radiographic Inspection of Metals.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Inspection Terms and Definitions.

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for percent Defective.

MIL-STD-1167 — Ammunition Data Card.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

7549097 — Cartridge, 106MM, HEAT, M344A1, w/ Fuze PIBD, M509A1

7548962 — Container, Ammunition, Fiber, M316 for Cartridge 106-MM, HEAT, M344-

FSC 1315

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A1 w/Fuze PIBD M509 for Rifle M40A1, Assembly, Details and Packing.

7548963 — Box, Packing, Ammunition, for 106MM Rifle Cartridges in Fiber Container M316, Assembly Details and Packing.

PUBLICATIONS

ARMY

IEL-7549097 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement and functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *General.* Materials shall be in accordance with applicable specifications and drawings.

3.1.2 *Clean scrap.* Clean scrap produced in regular operations may be remelted and reused provided it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.4).

3.2 **Cartridges.** The cartridges shall comply with all requirements specified on Drawing (dwg.) 7549097 and with all requirements specified in applicable specifications.

3.3 Cavitation.

3.3.1 *General.* Cavities having a maximum (max.) projected dimension of 1/32 inch or less shall be disregarded provided that they are separated by ½ inch of solid explosive.

3.3.2 *Maximum projected areas and dimensions of cavities.* The maximum projected areas and dimensions of cavities in each of the segments (see figure 1) shall be in accordance with table I.

TABLE I

Segment	Sum of areas of all cavities (square inches)	Maximum dimension of any cavity (inch)
A	⅛	⅜
B	⅜	⅝
C	¼	

3.3.3 *Cavities extending into two segments.* The projected area of cavities extending into two segments shall meet the requirements of both segments.

3.3.4 *Cracks and annular rings.* Cracks shall be permitted if they are no wider than 1/32 inch. The presence of annular rings 1/16 inch thick, max., and extending from the shell wall ¼ inch into the charge shall not be cause for rejection, except in Segment C. No annular rings are permissible in Segment C. The cavity area represented by such a crack or annular ring shall not be included when determining cavity area in the charge.

3.3.5 *Piping cavity.* A piping cavity on the longitudinal axis of the projectile between the apex of the cone and the faced surface of the charge shall be permitted provided that its projected length does not exceed 1½ inches and its width does not exceed ⅓ inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.3.6 *Surface cavitation.* The maximum dimension of any visible cavity in the faced surface (see 6.6) of the charge shall not exceed 3/16 inch. Individual cavities formed due to chipped or broken explosive from the side-

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walls of the fuze well shall not extend in aggregrate around more than ¼ of the fuze well circumference (1 inch). The total area of all cavities in the base of the fuze well shall not exceed ¼ square inch. The total area of all cavities in the rear face of the charge shall not exceed ¼ square inch.

3.4 Propellant weight. The weight of the propellant shall not differ by more than plus or minus 0.25 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.5 Bullet pull. The cartridge shall separate from the projectile at a load between 3000 pounds minimum (min.) and 7000 pounds max., when tested as specified in 4.3.4.

3.6 Electrical circuit resistance. The loaded and fuzed projectile, prior to assembly into a complete round, shall have an electrical circuit resistance of not less than 62,000 ohms nor more than 300,000 ohms when tested as specified in 4.3.2.

3.7 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.3.5.

3.7.1 *Muzzle velocity.* The corrected muzzle velocity at ambient temperature shall be not more than 1700 feet per second (f.p.s.) nor less than 1,600 f.p.s. and the standard deviation for the lot shall be not more than 11.0 f.p.s.

3.7.2 *Functioning and penetration.* The round shall function and there shall be complete penetration of 6 inch homogeneous armor plate at a 60 degree position from the vertical (see 6.5).

3.7.3 *Fin opening.* The fins shall open.

3.7.4 *Premature bursts and security of parts.* There shall be no premature burst or indication of metal parts separation or breakup in the rifle bore or in flight before impact.

3.8 Assembly torque. Threaded joints shall withstand subsequent re-application of the minimum assembly torque specified on the applicable drawings, without moving. These tests shall be conducted in accordance with 4.3.3.

3.9 Workmanship.

3.9.1 All parts shall be free of cracks, chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.9.2 The painting and marking shall be restored by the assembly contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated.

3.9.3 *Loading of high explosive.* The interior surface of the projectile shall be dry at the time the high explosive is poured. Solid explosive shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.9.4 After loading of high explosive the exterior of the projectile, the threads and the fuze cavity shall be cleaned to remove extraneous explosive and foreign material. If a chemical cleaner is required, no material other than acetone shall be used. Care should be taken to prevent acetone from coming into contact with the charge and becoming entrapped in the projectile.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Govern-

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ment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.4.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government shall not be cause for changing the lot number. Lots shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235, when applicable). Each lot shall contain:

(a) Cartridge cases of one type of metal and of one lot interfix number from one manufacturer in accordance with the same drawing and drawing revision, same specification and specification revision.

(b) Projectile metal parts from one manufacturer in accordance with the same drawing and drawing revision, same specification and specification revision.

(c) Propellant from not more than one lot.

(d) Loaded fuzes of one lot interfix number from one manufacturer.

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(e) Loaded primers of one lot interfix number from one manufacturer.

(f) Composition B of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Inspection for critical defects (and major defects when so specified) shall be 100 percent. Sampling plans and procedures for major and minor defects shall be in accordance with MIL–STD–105. Continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQLs and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each minor defect and an AQL of 0.40 percent for each major defect except when 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 **Projectile, prior to loading** (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Foreign matter in charge cavity	Visual	01001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	01002

4.2.2.2 Projectile, prior to assembling chamber (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Depth to face of charge	Gage	02001
2.	Depth of fuze cavity	Gage	02002
3.	Charge shock pad missing	Visual	02003
4.	Packing O-ring missing	Visual	02004
5.	Explosive on body threads	Visual	02005
Major: None defined			
Minor:	AQL 1.0 percent		
201.	Foreign matter in fuze cavity	Visual	02006
202.	Surface cavitation excessive (see 3.3.6)	Visual	02007

4.2.2.3 Projectile, prior to assembling fuze (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Fuze shock pad missing	Visual	03001
2.	Chamber fails to withstand min. assembly torque (see 3.8)	Test	03002
Major: None defined			
Minor: None defined			

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4.2.2.4 Projectile, prior to assembling fuze lock plug (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Fuze missing	Visual	04001
Major:	AQL 0.65 percent		
101.	Terminal screw obviously loose	Manual	04002
102.	Wire terminal or wire assembly damaged or loose	Visual-Manual	04003
103.	Fuze insulator missing or mislocated	Visual	04004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	04005

4.2.2.5 Filler (see dwg. 8837084 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Thickness from apex to base	Gage	05001
202.	Width of flat	Gage	05002
203.	Evidence of poor workmanship (see 3.9)	Visual	05003

4.2.2.6 Gas check (see dwg. 8837083 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Minor: None defined			
Minor:	AQL 1.5 percent		
201.	Wall thickness	Gage	06001
202.	Thickness from apex to base	Gage	06002
203.	Evidence of poor workmanship (see 3.9)	Visual	06003

4.2.2.7 Projectile, prior to assembling and caulking gas check (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Plug fails to withstand min. assembly torque	Test	07001
2.	Filler missing	Visual	07002
Major: None defined			
Minor: None defined			

4.2.2.8 Projectile, prior to assembling fin assembly (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Gas check missing or not caulked	Visual	08001

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Major: None defined

Minor:	Defects	Method of inspection	Code No.
	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	08002

4.2.2.9 Projectile, prior to assembling ogive cap (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Lucky assembly missing	Visual	09001
Minor: None defined			

4.2.2.10 Projectile, prior to assembly into cartridge case (see dwg. 7549096 covering a detail of dwg. 7549097).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Fin assembly fails to withstand min. assembly torque (see 3.8)	Test	10001
102.	Ogive cap fails to withstand min. assembly torque (see 3.8)	Test	10002
103.	Any component missing or damaged	Visual	10003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	10004

4.2.2.11 Primer, prior to assembling into case (see dwg. 8838009 covering a detail of dwg. 7549097).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Flash hole missing	Visual	11001
Major: None defined			
Minor: None defined			

4.2.2.12 Liner, prior to assembling into case (see dwg. 7549065 covering a detail of dwg. 7549097).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.0 percent		
201.	Liner damaged	Visual	12001
202.	Evidence of poor workmanship (see 3.9)	Visual	12002

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4.2.2.13 Cartridge case and liner, prior to assembling projectile (see dwg. 7549097).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Primer obviously loose	Manual	13001
Major:	AQL 0.40 percent		
101.	Foreign material in case	Visual	13002
Minor:	AQL 1.5 percent		
201.	Length to case dimple	Gage	13003
202.	Evidence of poor workmanship (see 3.9)	Visual	13004

4.2.2.14 Cartridge assembly (see dwg. 7549097).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Primer above flush	Gage	14001
2.	Liner damaged to extent that propellant can escape	Visual	14002
Major:	AQL 0.65 percent		
101.	Loading plug fails to withstand disassembly torque (see 4.3.3)	Test	14003
102.	Loading plug above flush	Gage	14004
103.	Liner damaged to extent that moisture barrier is broken but not to extent that propellant can escape	Visual	14005
Minor:	AQL 1.5 percent		
201.	Exterior coating damaged with an aggregate of bare spots totaling in excess of ¼ square inch	Visual	14006
202.	Marking misleading or unidentifiable including incorrect color code	Visual	14007
203.	Adhesive on bourrelet	Visual	14008
204.	Evidence of poor workmanship (see 3.9)	Visual	14009

4.2.2.15 Unsealed fiber container (see dwg. 7548962).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Packing component missing or out of place	Visual	15001
Minor:	AQL 0.65 percent		
201.	Container cap cannot be removed by hand	Manual	15002
202.	Cartridge cannot be removed by hand	Manual	15003
203.	Glue or asphalt on cartridge	Visual	15004

4.2.2.16 Sealed fiber container (see dwg. 7548962).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Container cut or damaged through all layers	Visual	16001
102.	Metal end loose or distorted	Visual-Manual	16002
103.	Tape incomplete or badly wrinkled	Visual	16003

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Minor:	AQL 2.5 percent		
201.	Tear tab length min	Gage	16004
202.	Content loose	Manual	16005
203.	Gap between cover and body of container exceeding ⅛ inch	Visual	16006
204.	Cuts, scuffs or gauges in outer layers	Visual	16007
205.	Marking misleading or unidentifiable	Visual	16008

4.2.2.17 Sealed wood packing box (see dwg. 7548963).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Box damaged exposing contents	Visual	17001
102.	Board loose or broken	Visual-Manual	17002
103.	Hardware or strapping missing, broken or loose	Visual-Manual	17003
104.	DOD symbol missing or incorrect	Visual	17004
Minor:	AQL 1.5 percent		
201.	Handle missing or insecure	Visual-Manual	17005
202.	Car seal missing, unsealed or improperly positioned	Visual	17006
203.	Marking misleading or unidentifiable	Visual	17007

4.3.3 *Testing.*

4.2.3.1 *Check test for possible deterioration of primers.* If the total elapsed time between original acceptance of the primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable primer specification immediately before the primer lot is assembled into the cartridges The test shall be performed by the contractor on assemblies selected by the Government Inspector at the facility assembling the complete cartridges (see 6.7).

4.2.3.2 Cavitation (see 3.3 except 3.3.6). Cavitation defects are classified as follows:

		Code No.
Critical:		
1.	Cavitation, cracks or annular rings in Segment A in excess of that permitted by the applicable requirement	02008
2.	Cavitation, cracks or annular rings in Segment B in excess of twice that permitted by the applicable requirement	02009
Major:		
101.	Cavitation, cracks or annular rings in Segment B in excess of that permitted by the applicable requirement, but not more than twice that permitted	02010
102.	Cavitation, cracks or annular rings in Segment C in excess of that permitted by the applicable requirement	02011
103.	Piping cavity in excess of applicable requirement	02012

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The sampling plan and procedure for this test for both critical and major defects shall be in accordance with MIL–STD–1235 using a CSP–2 plan having an AQL of 0.65 percent, except that when a critical defect is found, 100 percent inspection for i successive defect free units (for both criticals and majors) shall be required immediately instead of continuing on a sample basis. When a major defect is found during the f phase of the plan, inspection shall continue at the same frequency until the homogeneity requirement is fulfilled or 100 percent inspection is required. Appropriate i and f values shall be as specified by the procuring activity. The test shall be performed as specified in 4.3.1 with equipment in accordance with 4.2.4.

4.2.3.3 Projectile electrical test (see 3.6), Critical defect, Code No. 08003. This test shall be performed 100 percent. Any assembly that fails to comply with the applicable requirement shall be classed defective and removed from the lot. Fuze circuit resistance of 1,000 to 10.000 ohms is an indication of an armed fuze and appropriate safety precautions shall be taken. The test shall be performed as specified in 4.3.2, using equipment in accordance with 4.2.4.

4.2.3.4 Propellant weight (see 3.4) Major defect, Code No. 14010. The propellant shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing and, if performed manually, shall be performed by another operator. Any propelling charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot subject to correction prior to resubmission. Equipment shall be in accordance with 4.2.4.

4.2.3.5 Profile and alignment (see dwg. 7549097) Major defect, Code No. 14011. The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.6 Bullet pull (see dwg. 7549097) Major defect, Code No. 14012. The sampling plan for this test shall be in accordance with Standard MIL–STD–414. Section B, Part II, table B–3, Code Letter F and an AQL of 1.5 percent shall be used. The test shall be performed as specified in 4.3.4. with equipment in accordance with 4.2.4.

4.2.3.7 *Proving ground.* The following tests shall be performed as specified in 4.3.5.

4.2.3.7.1 Muzzle velocity (see 3.7.1) Major defect, Code No. 14013.

4.2.3.7.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, the sampling plan for this test shall be in accordance with Standard MIL–STD–414. Section B, Part II, table B–3, Code Letter I and an AQL of 1.0 percent shall be used. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.75 exceeds the requirement specified in 3.7.1, the lot shall be rejected.

4.2.3.7.1.2 After three consecutive lots have met the criteria of 4.2.3.7.1.1, the sampling plan and procedure for this test shall be in accordance with Standard MIL–STD–414. Section B, Part II, table B–3, Code Letter F and an AQL of 1.0 percent shall be used. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.65 exceeds the requirement specified in 3.7.1, the lot shall be rejected.

4.2.3.7.2 Functioning and penetration (see 3.7.2) Major defect, Code No. 14014.

4.2.3.7.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement, twenty-five sample cartridges shall be selected from each lot for this test.

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If three or more sample cartridges fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.7.2.2 After three consecutive lots have met the criteria of 4.2.3.7.2.1, ten sample cartridges shall be selected from each lot for this test. If two or more sample cartridges fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.7.3 Fin opening (see 3.7.3) Major defect, Code No. 14015. This test shall be performed in accordance with 4.3.5.3. Acceptance and rejection shall be as follows:

Total No. rounds in ballistic sample (both phases)	*Accept*	*Reject*
50	2	3
35	1	2
20	1	2

4.2.3.7.4 Premature bursts and security of parts (see 3.7.4) Critical defect, Code No. 14016. If any of the sample cartridges used in the proving ground tests fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.7.5 *Refiring.* If for any reason, the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional sample rounds as required.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL–7549097 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of cavities.* The loaded projectile shall be subjected to radiographic examination. The X-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the X-ray negative shall be made to determine compliance with applicable requirements. (Note—Radiographic equipment, operators and procedures shall be qualified in accordance with MIL–R–11470). In addition, the sensitivity of X-ray technique shall be 2.0 percent as determined in accordance with 4.3.1.1 and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.3.1.2. In the event an original X-ray picture leaves doubt as to acceptability, the projectile shall be re-X-rayed at an angle 90 degrees from that of the original exposure and in the same plane.

4.3.1.1 *Check of sensitivity of X-ray technique.* The sensitivity of X-ray technique

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shall be determined by the means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the image of the penetrameter on the X-ray film. The sensitivity of X-ray technique shall be considered satisfactory when the image of the penetrameter including the hole in the center, is sharply defined on the X-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thicknesses and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter (see MIL-R-11471). The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated by figure 2 of this specification.

4.3.1.2 *Check of photographic density of film.* The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding so that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards be used (see 6.7). Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

4.3.2 *Electrical resistance of projectile.* This test shall be conducted after assembly of fuze and fuze lock plug. The resistance shall be measured between the eyelet and ogive. The short circuit current of the test equipment shall not exceed 20 micro amperes.

4.3.3 *Torque tests.* The torque shall be applied while holding the parts immediately adjacent to the joints being tested so as to avoid possible loosening of other joints. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discerable movement between adjacent parts shall be regarded as a torque test failure.

4.3.4 *Bullet pull.* The sample cartridges shall be tested for security of crimp of the cartridge case by applying the load in an axial direction. The rate of travel of the testing machine head shall be .100 to .125 inch per minute, inclusive.

4.3.5 *Proving ground.* These tests shall be conducted at a Government proving ground in accordance with the applicable Ordnance Proof Manual and Acceptance Test Procedure. The cartridge shall be fired in a rifle for which it is standard and observation shall be made for compliance with all applicable requirements.

4.3.5.1 *Penetration.* Cartridges tested for penetration shall be fired at a 6 inch homogeneous armor plate placed at a 60 degree angle position from the vertical and approximately 400 feet from the muzzle of the rifle.

4.3.5.2 *Accuracy (for information only).* The cartridges fired in the muzzle velocity phase shall be fired at a screen placed 1,000 meters from the muzzle of the rifle. The hit pattern results shall be recorded on the firing record for informational purposes.

4.3.5.3 *Fin opening.* The rounds fired in the penetration and velocity accuracy tests shall be fired through $\frac{1}{16}$ inch chipboard or with photographic coverage to determine proper fin opening.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The cartridges shall be pre-

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served and packaged in accordance with dwg. 7548962.

5.2 Packing.

5.2.1 *Level A.* The cartridges in containers shall be packed in accordance with dwg. 7548963.

5.3 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL–STD–1167.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and nonmandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descrip-

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tive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contactors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be nonreproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 **Clean scrap.** Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in pouring and cooling operations, and scrap formed in the drilling operation.

6.5 **Penetration.** Complete penetration is defined as a hole completely through the plate and free from obstructions. However, the lodging of a copper slug in the completed hole is not considered cause for rejection.

6.6 **Faced surface.** The faced surface of the charge includes the bottom and sidewalls of the fuze well and the rear face of the charge.

6.7 **Cost of check test.** The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the perfomance of the check tests. The test shall be conducted at Govenment expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., per-

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mitted while the primer lot was under the jurisdiction of either contractor.

6.8 **Photographic densities.** "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N. Y., has been found satisfactory as a source of information on filter densities.

6.9 **Submission of test data and ammunition data cards.** In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer

ATTN: SMUPA-ND2
SMUPA-DC8

Picatinny Arsenal

Dover, N. J.

6.10 **Standardization agreement.** Certain provisions of this specification are the subject of international standardization agreement ABC ARMY STD-46A. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Department Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1315-A440

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